

Arsalan Ghogari

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Education

University of Southern California: Viterbi School of Engineering, Marshall

Aug 2023 - Present

School of Business

Bachelor of Science, Computer Science/Business Administration; Minor, AI Applications

Study Abroad: University of Bristol, UK | Spring 2026

- GPA: 3.9; Presidential Scholar, Viterbi Scholar, Dean's List, W.V.T. Rusch Honors, Thematic Option Honors
- Relevant Coursework: Data Structures & Algorithms, Organizational Leadership, Communications, Business Statistics, App Development

Experience

Director of Digital Strategy, USC Undergraduate Student Government

May 2025 - Present

- Manage and maintain USC USG's official WordPress website, improving accessibility, information architecture, and usability for over 20,000 students.
- Developed front-end features using HTML, CSS, and JavaScript, including dynamic homepage announcement systems and interactive election interfaces.
- Built and deployed [Legislative Project Tracker](#) and [Senate Bill Tracker](#) systems by integrating Google Forms, Sheets, and NinjaTables to automate transparency workflows.
- Developed an interactive [Funding Tracker](#) visualizing budget allocation across research, arts, philanthropy, and student initiatives.
- Built and deployed [Ask USG](#), a production RAG chatbot for USC's 20,000+ undergraduate student body: Express/Node + OpenAI on Render, Postgres/pgvector retrieval, SSE streaming widget integrated with the USG WordPress site.
- Designed a deterministic content-freshness gate that stamps every knowledge chunk with its WordPress last-modified date and routes users to staff when sources predate the current year, plus a mental-health crisis pathway (keyword + moderation detection, guardrailed scripted responses) routing to the 988 Lifeline.
- Resolved complex CSV parsing and synchronization issues affecting tracker reliability and maintainability.
- Invited (and accepted) to continue serving in the role for an additional academic year in recognition of technical contributions, initiative, and impact on USC USG digital infrastructure.

Perception & Autonomy Lead / Software Engineer, USC RoboSub

Jan 2024 - Present

- Lead development of the AUV's perception and autonomy stack, integrating computer vision, navigation, SLAM, and mission control for autonomous underwater operation.
- Collected and annotated underwater datasets to train YOLO26 and RT-DETR object detection models, improving detection performance from 0.40 to 0.70 to 0.90 mAP.
- Integrated vision with ROS-based SLAM by publishing perception outputs as ROS topics, enabling real-time localization and navigation using visual detections.
- Developed mission logic and navigation software across the autonomy stack, coordinating perception, planning, and control between ROS nodes.
- Built high-fidelity NVIDIA Isaac Sim environments modeling the AUV, cameras, and sensors with ROS integration, enabling closed-loop sim-to-real testing of the perception and navigation stack prior to in-water deployment.
- Recorded, managed, and analyzed ROS bag datasets in Foxglove to train and debug perception/navigation models, visualize sensor and detection topics, measure latency and detection metrics, and replay runs for reproducible, simulation-driven regression testing.
- Containerized ROS services with Docker and Kubernetes for deployment across NVIDIA Jetson Xavier and Raspberry Pi hardware.
- Built and maintained the team website using React, TypeScript, & Notion API, improving club visibility.

Math Peer Mentor, C.E.N.T.R.I.C. Program (USC Viterbi & Dornsife)

Aug 2025 - Dec 2025

- Supported launch of USC Viterbi and Dornsife's mentoring initiative for first-generation and transfer students through

one-on-one mentorship and outreach.

- Collaborated with 30+ professors, advisors, and residential staff to coordinate outreach and increase student engagement.
- Managed mentee application tracking, mentor matching, scheduling, and communication workflows

Technical Projects

[USC USG AI Chatbot](#) | **Node/Express, Python, OpenAI, Postgres/pgvector, LangGraph, Langfuse, GitHub Actions**

- Built and shipped a production RAG chatbot for USC's Undergraduate Student Government, auto-ingesting all 97 pages/posts on the USG WordPress site into Postgres/pgvector (HNSW, idempotent upserts), refreshed weekly via GitHub Actions with zero manual steps.
- Added two-stage retrieval (20-candidate vector recall → LLM rerank), improving MRR from 0.91 to 0.98 on a 51-question golden set measured by a self-built eval harness (hit-rate, MRR, LLM-judge faithfulness) with Langfuse tracing.
- Cut time-to-first-token from ~6.2s to ~2.3s via SSE token streaming while keeping a mental-health crisis pathway whole-message and guardrail-checked ahead of every response.
- Extended into a tool-using agent (LangGraph state machine) choosing between KB retrieval and live data tools (events calendar, project tracker) at 100% tool-selection accuracy (22/22); hardened with rate limiting, CORS allowlist, input validation, Docker, and CI with an automated eval job.

[Parley — AI Voice Negotiation Agent](#) | **Next.js, TypeScript, ElevenLabs Agents (WebRTC), OpenAI GPT-4o, Supabase, Google Places API**

- Built solo for the ElevenLabs × Hack-Nation Global AI Hackathon: a voice agent that discovers local movers via live Google Places, phones them, extracts itemized quotes into strict-JSON, negotiates using competing bids as leverage, and delivers a ranked, evidence-backed report with transcripts and call recordings — then calls the winner back to book.
- Engineered dual-voice live calls (ElevenLabs WebRTC agent + TTS counterparty) with deterministic turn-taking — per-turn reply generation, keep-alive pings against agent turn timeouts — producing genuine mid-call price drops (\$2,400 → \$2,250 binding) from leverage the agent gathered, not scripted.
- Enforced honesty in code, not prompts: a per-turn tripwire parses every spoken dollar amount (digits and words), validates citations against job-scoped stored quotes, corrects the agent mid-call, and surfaces violations in the UI; an offline guardrail audit verifies red-flag, itemization-reconciliation, and citation rules over all stored data.

[AI-Powered Smart Seasoning Carousel](#) | **React, Tailwind CSS, OpenAI GPT-5.4, ESP32**

- Built an AI-powered smart seasoning carousel integrating an ESP32 microcontroller with a React/Tailwind interface to streamline cooking workflows and ingredient organization.
- Integrated GPT-5.4-based recipe generation to recommend personalized meals and seasoning combinations based on available ingredients and user preferences.

[Collaborative Note-Taking Application](#) | **Java, XML, Android Studio**

- Developed a collaborative Android note-taking application featuring intuitive UI workflows and persistent note management functionality.
- Selected as one of the top 2 implementations out of 37 teams and invited to present the project to the class for technical and design excellence.

Skills

Programming: Python, C++, C, Java, JavaScript, TypeScript, React, HTML/CSS, MATLAB

Technologies & Frameworks: ROS, Docker, OpenAI API, YOLO, WordPress, Java Servlets, Express/Node,

PostgreSQL (pgvector), LangGraph, Langfuse, GitHub Actions CI/CD, Next.js, ElevenLabs Agents, Google Places API

Tools: Git, Google Workspace, MS Office, OnShape, AutoCAD